

Optical Flow Analysis of the Aurora Borealis

Erik Mårten Blixt, Joshua Semeter, *Member, IEEE*, and Nickolay Ivchenko

Abstract—Optical observations of the aurora have traditionally focused on the structure, intensity, and wavelength of the emissions. But the apparent motion of auroral forms offers another important diagnostic tool for investigating this poorly understood phenomenon. Prior analyses of auroral motion have focused on tracing individual features. In this letter, we investigate the feasibility of deriving the entire two-dimensional velocity field automatically using robust optical flow estimation. The analysis is applied to two narrow-field video sequences. Both examples are rich in small-scale structure and motion, but appear very different to the eye. The robust optical flow estimator performed well for regions of dense turbulent motion, while sheared flow and flow which is perpendicular to image intensity gradients, was poorly resolved. The relative magnitude of the outliers provides a quantitative measure of the validity of the underlying flow model and, hence, a means of automatically differentiating among auroral forms with differing physical origins. The technique can be used to deduce ionospheric electric fields and neutral winds, and the flow fields yield important physical information about the generation mechanism.

Index Terms—Aurora borealis, optical flow, robust statistics.

I. INTRODUCTION

THE aurora borealis is created by collisions between charged particles from the magnetosphere (mostly electrons) and neutral gases in the Earth's upper atmosphere (mostly O₂, N₂, and O). The charged particles are accelerated into the atmosphere at an altitude of several thousand kilometers, and although our overall knowledge of the large-scale acceleration processes are steadily improving, the fine scales, and the variability, of the aurora are still not well understood [1], [2]. Because the charged particles are strongly tied to the magnetic field ($\mathbf{B}$), which is nearly vertical at high latitudes where aurora is typically observed, the aurora is often likened to a horizontal screen on which the particle flux is projected and displayed.

However, it remains uncertain what the apparent motion of the aurora really represents. If the projection interpretation is accurate, then auroral forms should drift with the convective circulation of the magnetosphere and, hence, with the ionospheric plasma (the $\mathbf{E} \times \mathbf{B}$ drift). Haerendel *et al.* [3], however, observed a relative drift between the aurora and the ionosphere of a few 100 m/s, suggesting that the magnetosphere and the ionosphere may become electrically decoupled in the

vicinity of auroral arcs. Far greater discrepancies have been observed for smaller scale auroral structure (e.g., folds and curls) which can drift orders of magnitude faster than the ionospheric plasma [4], [5]. These motions are thought to project plasma dynamics in the acceleration region [6]. Vogt *et al.* [7], for instance, studied whether the sheared flow of small-scale auroral vortices (so-called "curls") was consistent with an electrostatic acceleration model. Auroral motions have also been used to constrain possible generation mechanisms for black aurora [8], [9].

This letter describes an initial attempt to apply optical flow estimation to auroral image sequences. Prior studies involving auroral motion have relied on rather crude techniques for estimating auroral velocity. The most widely used technique is to simply track elemental features by hand [3], [8], [9]. A somewhat more sophisticated approach was applied in [7], who used a slice-stack principle to estimate one-dimensional cuts through the flow field. Optical flow analysis, on the other hand, provides a framework by which the entire two-dimensional flow field for an image sequence may be estimated. A reliable estimate of the flow field will provide access to the phase space of the aurora, which yields information about stationary points and their stability, vorticity, etc., important information for understanding the physical mechanism behind the aurora. Furthermore, most auroral classification schemes rely entirely on still images. In many cases, auroras that look similar in still images behave very differently when viewed as an image sequence. A flow field estimate can help quantify these spatio-temporal differences. Another potential application of the flow field is to use the drift of auroral features stemming from long-lived forbidden emissions to deduce ionospheric electric fields and neutral winds [10].

A wide variety of techniques have been developed to estimate the flow of objects in image sequences [11]–[13]. The unique characteristics of the aurora call for a robust estimation approach. The particular algorithm applied herein provides not only an estimate of the flow field, but an estimate of where the constraints in the motion model are violated. This latter information may have value as a classification scheme; violations in data conservation versus data smoothness may be useful for discriminating between auroras with different physical generation mechanisms.

II. OBSERVATIONS

The auroras studied in this letter were captured using the digital auroral imager Odin, belonging to University of Tromsø. The imager consists of a GEN III, extended blue, image-intensified (XYBION ISS-750) camera, with a F0.95 lens, $14.35 \times 10.92^\circ$ field of view. All observations were panchromatic, looking parallel to the magnetic field. The video stream was digitized in real time, and converted to digital video (25 f/s). A GPS-controlled timestamp allowed us to keep track

Manuscript received July 29, 2005.

E. M. Blixt is with the Faculty of Science, Auroral Observatory, Department of Physics, University of Tromsø, 9037 Tromsø, Norway and also with the Department of Computer and Electrical Engineering, Boston University, Boston MA 02215 USA (e-mail: marten.blixt@phys.uit.no).

J. Semeter is with the Department of Computer and Electrical Engineering, Boston University, Boston MA 02215 USA.

N. Ivchenko is with the Alfvén Laboratory, Royal Institute of Technology (KTH), SE-100 44 Stockholm, Sweden.

Digital Object Identifier 10.1109/LGRS.2005.860981

TABLE I
OVERVIEW OF EVENTS ANALYZED IN THIS LETTER

	Date & Time [UT]	Location
A	2004-02-23, 03:49:00–03:52:30	Søndrestrømfjord, GRL
B	2003-11-18, 17:15:54–17:16:25	Tromsø, Norway

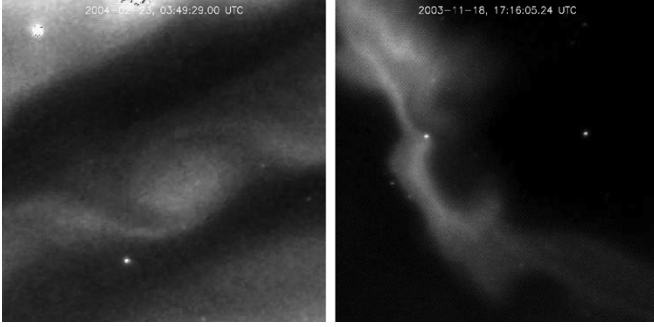


Fig. 1. (Left) Aurora from event A, which shows a curl at the center, and (right) from event B, which shows an arc-like feature which undergoes large deformations at very high speed.

of time for each frame. The lens yielded a 26.4×21.1 km view at the altitude of the auroras (≈ 105 km). Because the auroras are optically thin, and often are stretched out along the field lines, the apparent brightness will change toward the boundary of the image plane. This is an unwanted geometric effect, and we tried to minimize its impact by only looking at a $7.0 \times 7.0^\circ$ region at the center of each image, which corresponds to about a 12×12 km section of the aurora. Long-lived auroral emissions causes blurring when the aurora moves rapidly, and a trailing tail of light may contaminate the result. This effect can be reduced by filtering away the long-lived emissions. Usually, this is done by applying a long-pass filter at 630 nm. This has not been done in this study. In the first event, the motions are fairly slow, so the contamination is not expected to be large. In the second event, the obvious fast motions are in themselves a proof that prompt emissions are dominating.

The time and location of the events studied in this letter are presented in Table I. Event A is characterized by auroral curls forming and disappearing about 2 h after local magnetic midnight during an active period. This event was chosen because curls are thought to represent a Kelvin–Helmholtz instability in a magneto-hydro-dynamics (MHD) plasma, and thus should have a fluid-like motion that is more likely to be well described by our basic motion model. By contrast, event B is characterized by small-scale (subkilometer) structures moving with speeds up to 150 km/s. In many instances, the structures appear to form in place, stepping through the field rather than flowing. This event occurred about 5 h prior to magnetic midnight, during the onset of a magnetic substorm. The fast, discontinuous motion of these auroras makes them less likely to be well described by our underlying optical flow model. Fig. 1 shows a snapshot of the aurora from each event.¹

¹ Video sequences that show the aurora can be downloaded in *avi* format at <http://www.phys.uit.no/~maarten/Results/OpticalFlow05.html>.

III. ROBUST OPTICAL FLOW TECHNIQUES

Optical flow is the representation of the *apparent* two-dimensional motion of an “object” (whose proper three-dimensional motion is described by what is called the *motion field*) projected on the image plane [14], [15]. We follow the method described by Black and Anandan [16], who implement a robust version of the regularization technique for the gradient-based method first proposed by Horn and Schunck [14]. Robust estimation is becoming an important tool in computer vision, as data rarely come from one single statistical distribution [17], and the gradient method has been shown to perform well when compared to other methods [12], [13]. In short, the flow that best describes the apparent motion on the image plane is found through a regularization technique that minimizes the error of the calculated flow relative to a set of models that constrains the flow [12]–[14], [18].

The standard model that forms the basis for many optical flow estimates is the *data conservation constraint*, or *brightness constraint equation* [16], [19], which simply assumes that the brightness (data) is conserved along a path $\mathbf{x}(t)$ in the image sequence

$$\frac{DI(\mathbf{x}(t), t)}{Dt} = \nabla I \cdot \mathbf{u} + \frac{\partial I}{\partial t} = I_x u + I_y v + I_t = 0 \quad (1)$$

where I is the brightness (or intensity or data) and $\mathbf{u}$ is the velocity $[u, v] = [\partial x / \partial t, \partial y / \partial t]$. Thus, according to the model, any change in brightness in the image sequence is solely due to motion of objects. The two most important reasons why the chosen model might not fit the auroral data perfectly are: 1) changes in the causative particle flux may result in variations in luminosity and shape and 2) because the aurora is optically thin, its apparent brightness will change with position of the features in the image plane as the path length through the feature changes. One fairly simple improvement, so that the model will become less sensitive to changes in auroral shape and size, is to include the compressible $(\nabla \cdot \mathbf{u})I$ term in (1) [20].

Equation (1) has two unknowns (u and v); we need at least one more constraint to yield a well-posed problem. The standard practice is to use a *spatial coherence constraint*, which assumes the flow is smooth in a local neighborhood. This smoothness constraint is often formulated as a *first-order* or *membrane* model [14], [16], which in a standard least square norm will yield an error

$$E_S(u, v) = \left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 + \left(\frac{\partial v}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial y} \right)^2 \quad (2)$$

which is to be minimized.

Because the model constraints are likely to be violated at some points, we have chosen to use a robust estimator, which ideally should ignore these outliers. There are two main reasons for outliers—failure of the motion model, or noise [18]. In robust statistics there is no *a priori* assumption that the noise is normally distributed, and outliers are down-weighted in the fitting procedure [21], [22]. Furthermore, the locations of the outliers—i.e., where the motion model is violated—can be identified [16], [23], [24].

There are some important benefits to using robust estimation in the auroral optical flow problem. First, if the region used to estimate the image gradient contains multiple motions, the solution will be fitted to the majority of the data rather than the average. A second, more interesting, consequence is that if the motion of the aurora does not follow our motion model (i.e., conservation of data along the path of motion, and smooth variations in velocity) the robust estimator will identify these regions. We may use this information to study how the robust optical flow technique treats different kinds of auroras, including its potential use as a physical classification strategy.

Following the example of [16] we use the *Lorentzian* error norm given by

$$\rho(x, \sigma) = \log \left(1 + \frac{1}{2} \left(\frac{x}{\sigma} \right)^2 \right) \quad (3)$$

where σ is a scale parameter which determines the width of the norm, i.e., the residual x is an outlier whenever $|x| \geq \sqrt{2}\sigma$. The value of the parameter σ is chosen to suit prior expectations of motion discontinuities and noise, but the solution is fairly stable to changes in σ of less than an order of magnitude [16]. We have used $\sqrt{2}\sigma_S = 0.05$ pixels per frame to represent a motion discontinuity (spatial coherence), and a more conservative estimate of $\sqrt{2}\sigma_D = 5$ for the data conservation term. The latter estimate is found through the variance of the residual of the data conservation $I_x u + I_y v + I_t$, at locations where the flow is zero. This means that σ_D is more or less determined by the variance in $\partial I / \partial t$ [11].

The total error consists of two components, namely the error in the data conservation (D) and spatial coherence (S) constraints

$$\begin{aligned} E(u, v) &= E_D(u, v) + E_S(u, v) \\ &= \sum_{s \in N} [\rho(I_x u_s + I_y v_s + I_t, \sigma_D) \\ &\quad + \lambda \left(\sum_{n \in G} \rho(u_s - u_n, \sigma_S) \right. \\ &\quad \left. + \sum_{n \in G} \rho(v_s - v_n, \sigma_S) \right)] \quad (4) \end{aligned}$$

where λ is a weighting parameter (set to 1/10 in this study) and G is the local north, south, east, and west neighbors of the point s . The minimization is carried out over the whole image (N). This error is minimized using a *graduated nonconvexity* method, which means that the scale parameter σ is initially set large enough to treat all residuals as inliers. The scale parameter is then gradually decreased toward its final value, so that the global minimum is properly found. To handle motion larger than single pixels, a coarse-to-fine strategy is used [11], [16]. Minimization is first carried out on a subsampled coarse image, and the result is then projected to a finer level until a dense solution exists for each pixel in the image. This means that our method is capable of both detecting large-scale motions, like the translational motion of an arc, as well as the motion of smaller scale perturbations superposed on the arc (if detectable above the noise floor). The details of this procedure are described in [16].



Fig. 2. Results from event A. (Left) The velocity field (the dot denotes the origin of the flow vector, and the length and direction of the line segment reflect the speed and direction of the flow; the length is here scaled with a factor of eight) shown atop an inverted image of the aurora. (Middle) The location where the data conservation constraint is violated (black areas). (Right) Shows where the smoothness constraint is violated (black areas).

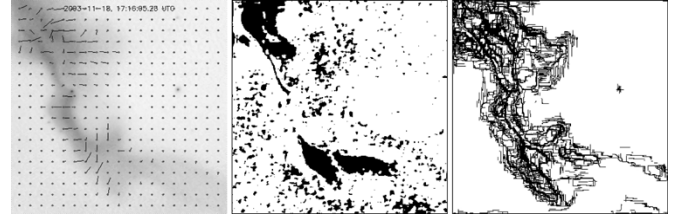


Fig. 3. Results from event B, otherwise same as Fig. 2. Here the velocity vectors are scaled with a factor of two.

IV. ANALYSIS

Attached videos show the auroras for both events. Event A is shown in true speed, while event B is at half speed. The motion in event B was so large that it was necessary to *deinterlace* the images (each image consists of two consecutive fields made up from even and odd lines, respectively). The resulting optical flow field, and the identified outliers (one for the data conservation constraint and one for the smoothness constraint), has only been calculated for a limited 20-s period within each event (from 03:49:11.8 to 03:49:31.7 for event A and from 17:15:59.2 to 17:16:19.1 for event B), using the true speed (25 f/s) of the sequences. Movies that show these results can be downloaded at <http://www.phys.uit.no/~maarten/Results/OpticalFlow05.html>.

To reduce the influence of noise we have scaled each image so that gray levels below about two variances in image noise are ignored ($\lesssim 10$). Because the vortices in event A are weaker than the bright aurora at the upper limb of the images, we have chosen to ignore gray levels above 150 in that event. In this way, a better focus is put on the curls at the center of the image. To reduce the influence of noise, and an interference pattern present in the data from event B, the images have been filtered first with a band-reject filter and then with a median filter in a 3×3 neighborhood.

At each pixel in the image, the residuals in the data conservation and spatial coherence terms were compared with the respective scale parameters $\pm \sqrt{2}\sigma_{D,S}$. In this way the outliers and their location are identified. These locations are indicated by the black areas in Figs. 2 and 3. The downloaded movies show them in detail.

We are interested in the number of spatial and data outliers as a measure of how well the motion model describes the flow of the aurora. For this we have computed the relative number of outliers: the number of pixels that are considered outliers divided by N ($N = 371 \times 371$ is the total number of pixels). This

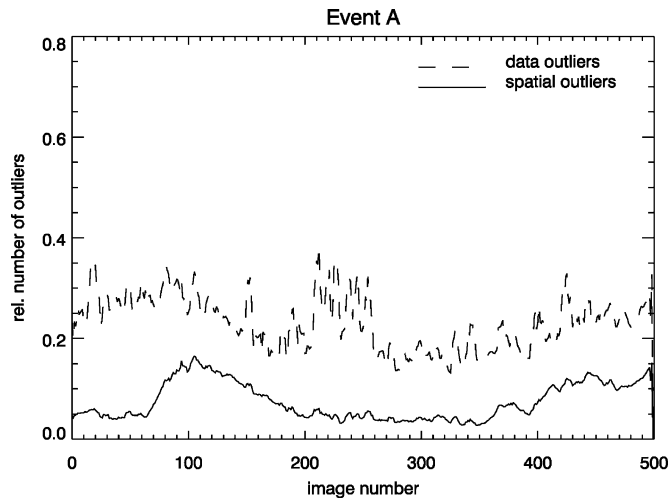


Fig. 4. Comparison of the relative number of data constraint violations (total number of outliers divided by the total number of pixels) and spatial coherence violations for event A.

is plotted versus image number (time) in Figs. 4 and 5, for the two events respectively.

V. DISCUSSION AND RESULTS

The attached flow field videos show the result of the optical flow analysis. Two snapshots are shown in Figs. 2 and 3. As can be seen, the flow is very irregular but describes the motion of the aurora fairly well. While it is desirable to have a quantitative measure of the errors in the estimated flow, unfortunately this is not possible for real image sequences when the motion of the objects are not known [12], [13]. Notice how the velocity vectors are mostly directly parallel to the intensity gradients. This is a direct consequence of the data conservation constraint. A solution solely based on this constraint would confine the optical flow to be along the image gradient, ∇I .

Results from event A are contaminated by noise, this is clearly seen in the resulting data outliers, who have a rather random appearance. The velocities are significantly smaller than in event B (the vectors in Fig. 2 are multiplied with a factor of four relative to those in Fig. 3), and there are moderate gradients present in the flow field. As can be seen in the movie sequence of the flow field, mostly the translational motion of the vortex is being picked up, but some of the vorticity is also detected, as can be seen in Fig. 2.

Event B exhibits much larger flow velocities (estimated up to 150 km/s at 100-km altitude in an example not shown), and shows many violations to both the data conservation and spatial coherence constraint. The violations are also seen to be more associated with auroral features, and not so noise-prone as in event A. In the middle of the analyzed sequence there is a period when almost no aurora is visible in the images. This period corresponds to the clear drop in relative number of outliers in the middle of Fig. 5. In the second half of the analyzed period, the aurora grows very bright so that the images are saturated. In saturated image areas and regions where there is very little structure, the optical flow technique fails completely. In this specific case, they correspond to the marked drops in data and spatial outliers seen at around image numbers 400 and 450 in Fig. 5.

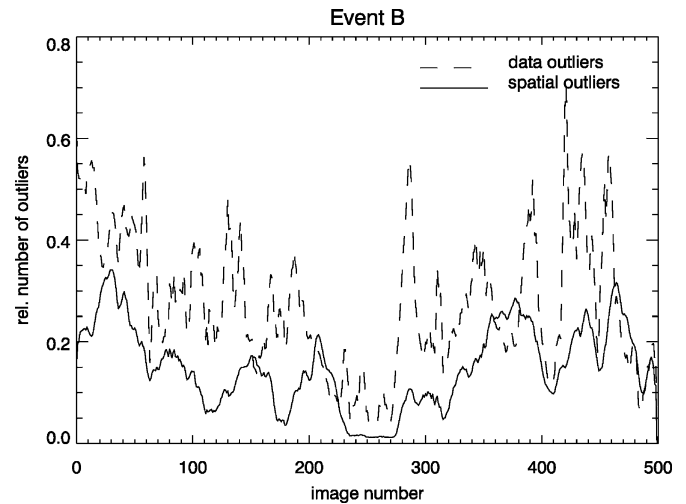


Fig. 5. Comparison of the relative number of data and spatial coherence violations for event B.

Despite the noise contamination in event A, the relative number of data outliers are, in general, lower in event A than in B (except in periods where there is no aurora visible in event B), as can be seen in Fig. 4 and 5. This suggests that the aurora in event A is better described by the standard motion model. Also, Figs. 4 and 5 show that the spatial coherence outliers are generally fewer than the data outliers, which has to do with the noise sensitivity of the data constraint.

The auroral motions in event A are fairly well understood [4]. The curl structure is thought to be generated by a negative charge sheet in the acceleration region, whose associated potential structure sets up a sheared $\mathbf{E} \times \mathbf{B}$ velocity which finally leads to a Kelvin–Helmholtz instability. The nature behind the aurora in event B is less well known, but it is not likely to be produced by a (quasi) stationary parallel electric field. Because the $\mathbf{E} \times \mathbf{B}$ velocity is incompressible ($\nabla \cdot \mathbf{v} = 0$) in most space plasmas, the simple motion model used in this letter is better suited to describe event A than B.

VI. CONCLUSION AND FUTURE WORK

There are some unique challenges in applying optical flow analysis to the aurora borealis which must be addressed before it becomes a standard tool in auroral research.

- The noise sensitivity should be decreased. This can be accomplished either by improving filtering techniques prior to the optical flow analysis, or simply by improving the effective signal-to-noise ratio of the imaging system.
- Shear flow and flow perpendicular to the image gradient are poorly represented because of the strong dependence on ∇I in the data conservation term. Careful application of a modified smoothness constraint, temporal constraints, or improved numerical differentiation [12] could improve this. Noise reduction also plays a crucial role in detecting flow perpendicular to ∇I .
- The basic motion model assumes that the intensity within an object is constant. This is clearly violated in a dynamic auroral display. One way to handle such brightness variation is to invoke a physical (linear) model of the variations

[19], and solve for these additional model parameters in the minimization process. This is hard to implement when the physical causes behind the brightness variations are less well understood, as in the aurora.

- The resulting flow field has a lot of variability from one image to the next. Using additional time-domain constraints, such as invoking more than two consecutive images at a time [23], may improve the stability of the solution.
- By using the generalized optical flow constraint [20], the tolerance for compressible flows and changes in auroral shape and area can be increased.

In conclusion, this letter presents an initial attempt at applying robust optical flow estimation on auroral image sequences. The resulting flow represents the motion of the aurora fairly well, but there is still room for many improvements in the estimation technique. The resolved flow field, and the accompanying constraint violation information, may be useful for auroral classification, phase space studies, and remote sensing of ionospheric electric fields and neutral winds.

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